Summary of Factor Correlation Analysis

1 Purpose

This analysis examines which factors explain the return behaviour of the AI equity basket and which factors are too similar to be used together. The factor set includes market, rates, credit, volatility, dollar, commodity, style, semiconductor, software, power, grid, and uranium exposures.

2 Factor Redundancy

Several factors contain almost identical information. The semiconductor factors `SEMI` and `SEMI_SOXX` are highly correlated. The two cyber-security factors, `CYBER` and `CYBER_HACK`, are also highly correlated. Software and cloud factors are closely related, and the rate factors `D10Y` and `SLOPE` overlap substantially.

Therefore, the full factor set should not be used directly in one regression. The full market-neutral factor set has a high condition number, indicating substantial multicollinearity. A smaller primary-factor set has much better diagnostics and is more appropriate for the final regression model.

3 AI Basket Results

The AI basket became more sensitive to the equity market during the holdout period. The market beta of the core basket increased from approximately 1.16 in the estimation period to approximately 1.74 in the holdout period.

Generic macro factors, such as rates, volatility, the dollar, and oil, add relatively little explanatory power beyond the market factor. Style factors improve the model more clearly. The largest additional improvement comes from the AI-adjacent power chain: power, grid, and uranium factors materially increase the out-of-sample explanatory power of the core basket.

For the core basket, the out-of-sample R^2 increases approximately as follows:

$$0.575 \rightarrow 0.579 \rightarrow 0.680 \rightarrow 0.780 \rightarrow 0.802.$$

These values correspond to the market-only model, the addition of macro factors, the addition of style factors, the addition of power/grid/uranium factors, and finally the addition of AI-close factors.

4 Interpretation

The results suggest that the AI basket is driven by three important sources of exposure:

- broad equity-market risk;
- style effects, especially the contrast between low-volatility and growth-oriented equities;
- the AI power-supply chain, including grid infrastructure, power generation, and uranium.

Semiconductor and software factors explain the chip and hyperscaler components strongly. However, these factors may partly overlap with the basket holdings themselves. Their explanatory power should therefore be interpreted as AI-equity exposure rather than as a fully independent economic driver.

The data-centre and neocloud segment is less stable out of sample. Coefficients estimated before 2024 do not transfer as reliably to this group, so rolling or expanding estimation windows are preferable.

5 Implication for the Final Model

The final model should use a compact set of non-redundant factors. A reasonable starting structure includes the market factor, one rates factor, one credit or volatility factor, one style factor, and selected power, grid, and uranium factors. Semiconductor and software factors should be included separately as an AI-close extension.

Because the basket was defined retrospectively, the results should be interpreted as historical factor attribution for a fixed AI basket, rather than as a fully investable real-time backtest.